

Week7HW-checkpoint

May 31, 2025

1 Week 7 Assignment

1.1 Problem 1

In the cell below, make a bar chart of the WSU tuition prices from 2012-2020.

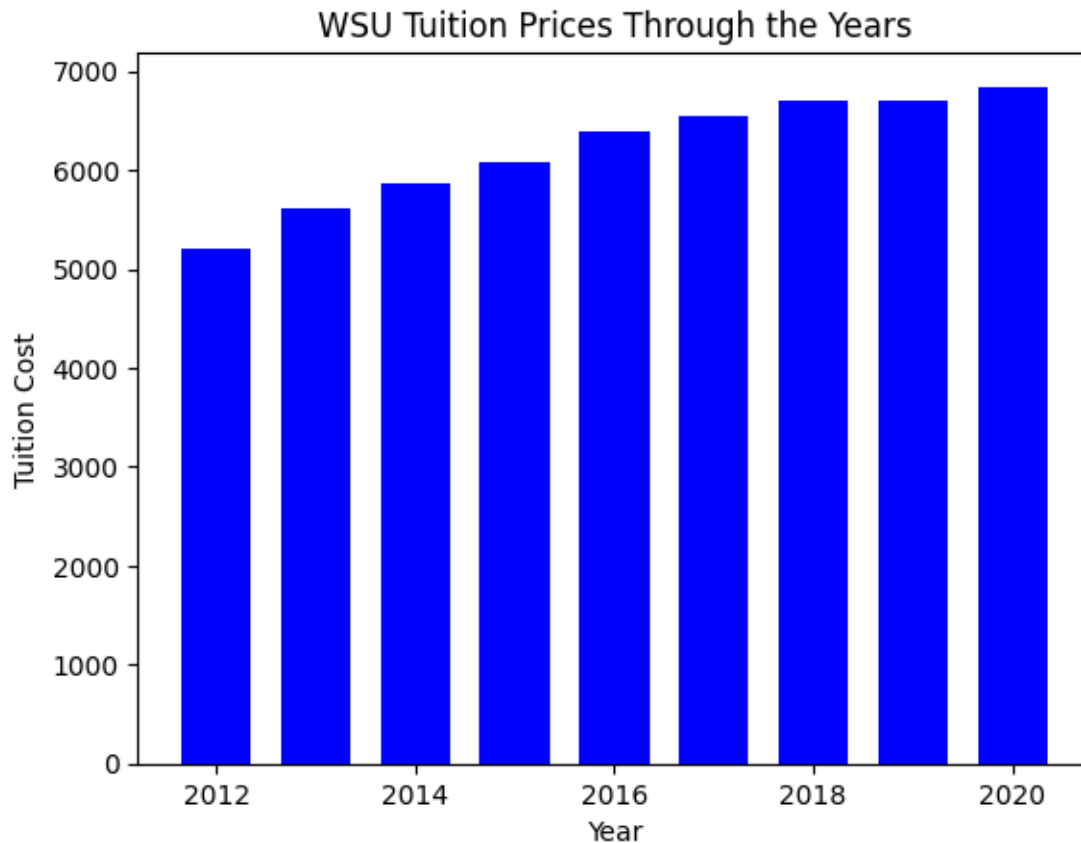
```
[1]: # 1. Imports
import matplotlib.pyplot as plt

# 2. Define Data
x = [2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020]
y = [5205, 5622, 5869.5, 6081, 6385.2, 6544.8, 6708.6, 6708.6, 6842.7]

# 3. Plot the bar
plt.bar(x, y,
        color='blue',
        width=0.7)

# 4. Label & Name Graph
plt.title('WSU Tuition Prices Through the Years')
plt.xlabel('Year')
plt.ylabel('Tuition Cost')

# 5. Show the plot
plt.show()
```



1.2 Problem 2

In the cell below make a bar chart and a pie chart from the `College_Rankings` data, showing the number of schools rated by state. Which one is easier to read?

After completing both the bar chart and the pie chart I would argue that the pie chart is easier to read and more userfriendly.

```
[42]: # Imports
import matplotlib.pyplot as plt
import pandas as pd

# Call College Ranking Data
College_Rankings = pd.read_csv("College_Rankings.csv")
#College_Rankings.head()

# Added to verify the names of the columns
print("Available columns in your DataFrame:")
print(College_Rankings.columns)
```

```

# Assigned values to x & y
x = (College_Rankings["Rank"])
y = (College_Rankings["State"])

# Write bar chart
plt.title("Number of Schools Ranked by State")
plt.bar(x,y,
        color='red',
        width=0.3)

plt.show()

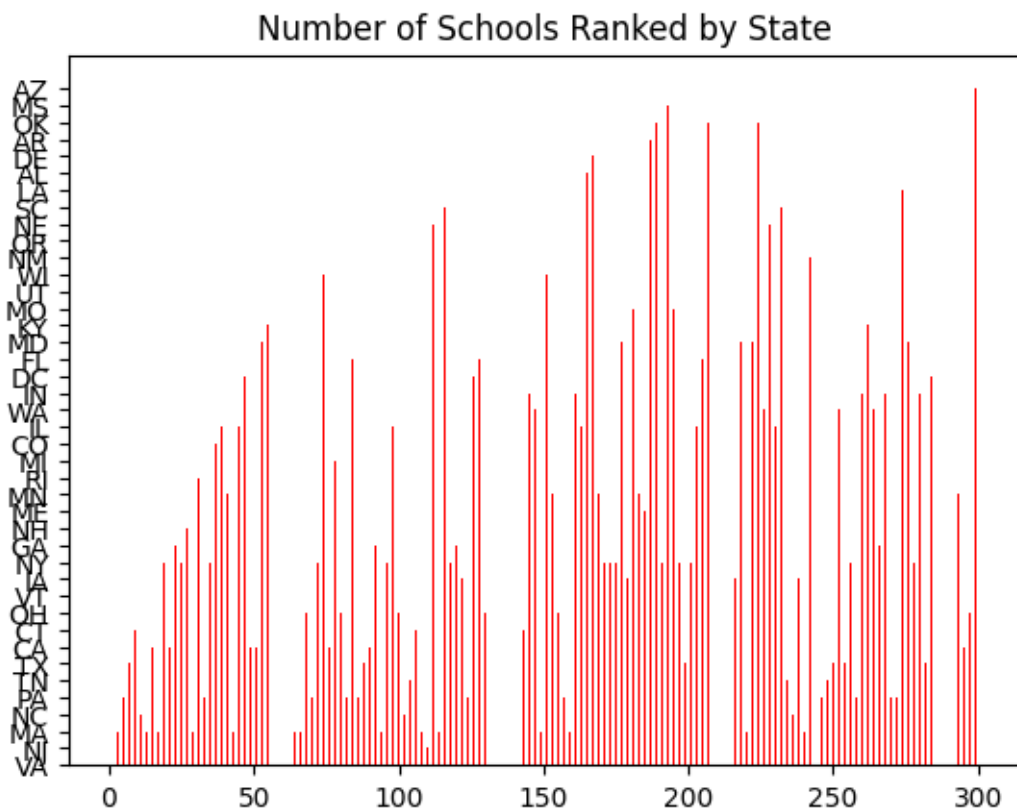
```

Available columns in your DataFrame:

```

Index(['Rank', 'Name', 'State', 'Type', 'Admit', 'Grad4', 'Cost', 'NeedAid',
      'NonNeedAid', 'NNApercent', 'GradDebt', 'Salary', 'Year'],
      dtype='object')

```



```

[49]: # Imports
import pandas as pd
import matplotlib.pyplot as plt

```

```

# Calls for College Ranking Data
College_Rankings = pd.read_csv("College_Rankings.csv")

# Count schools per state
schools_by_state = College_Rankings['State'].value_counts()

# Creating pie chart
plt.figure(figsize=(10, 10))

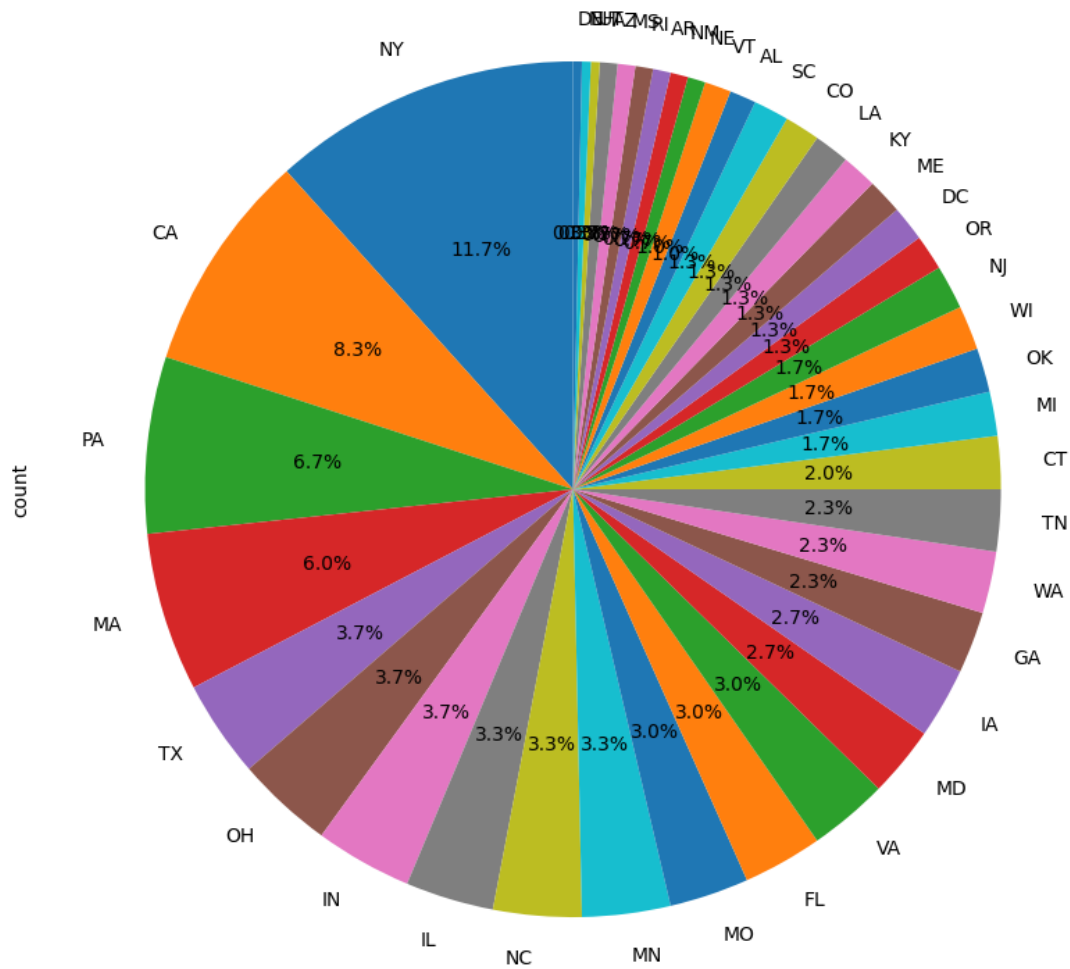
schools_by_state.plot(
    kind='pie',
    autopct='%1.1f%%', # Shows percentages on slices
    startangle=90,      # Starts the first slice at the top
)

plt.title('Number of Schools by Ranked by State')

plt.show()

```

Number of Schools by Ranked by State



1.3 Problem 3

In the cell below, create a histogram of the Gasoline prices in the COL data.

```
[ ]: # imports
import matplotlib.pyplot as plt
import pandas as pd

# Creating histogram of the Gasoline prices in the COL data
# Sample COL data with gasoline prices
data = {
    'City': ['Pasco', 'Kennewick', 'Richland', 'Seattle', 'Yakima'],
    'Gasoline_Price': [3.98, 4.13, 4.20, 4.60, 4.10]
```

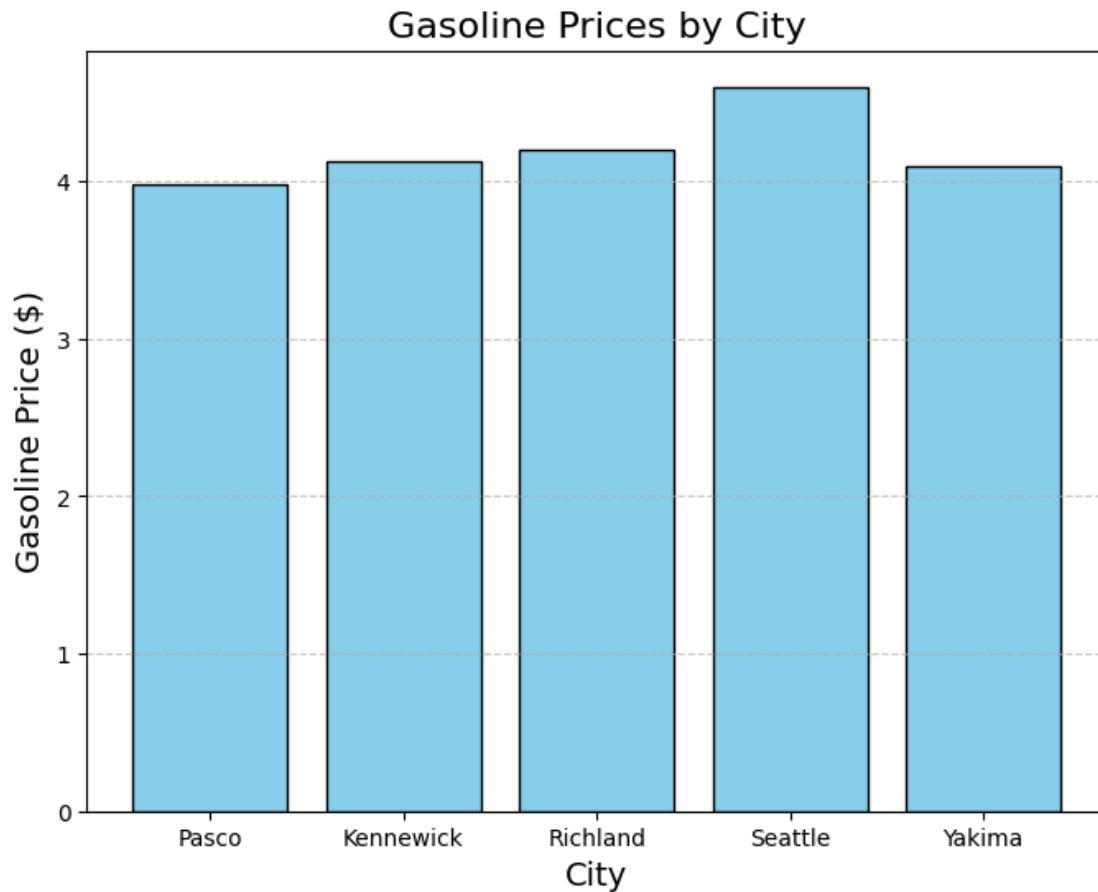
```

    'COL': []
}

# Convert data to a DataFrame
df = pd.DataFrame(data)

# Plot the bar chart of gasoline prices
plt.figure(figsize=(8, 6))
plt.bar(df['City'], df['Gasoline_Price'], color='skyblue', edgecolor='black')
plt.title('Gasoline Prices by City', fontsize=16)
plt.xlabel('City', fontsize=14)
plt.ylabel('Gasoline Price ($)', fontsize=14)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

```



1.4 Problem 4

In the cell below make a stacked histogram of the cost of the schools in the `College_Rankings` data, separated by the `Type` of school.

```
[ ]: import matplotlib.pyplot as plt
import pandas as pd

# Load your college ranking data
College_Rankings = pd.read_csv("College_Rankings.csv")

# 1. Get all the different types of colleges
school_types = College_Rankings['Type'].unique()

# 2. Create a list of 'Cost' data for each type of school.
costs_by_type = []
for s_type in school_types:
    type_data = College_Rankings[College_Rankings['Type'] == s_type]
    costs_by_type.append(pd.to_numeric(type_data['Cost'], errors='coerce').
↳dropna())

# --- Create the stacked histogram ---
plt.figure(figsize=(12, 7))

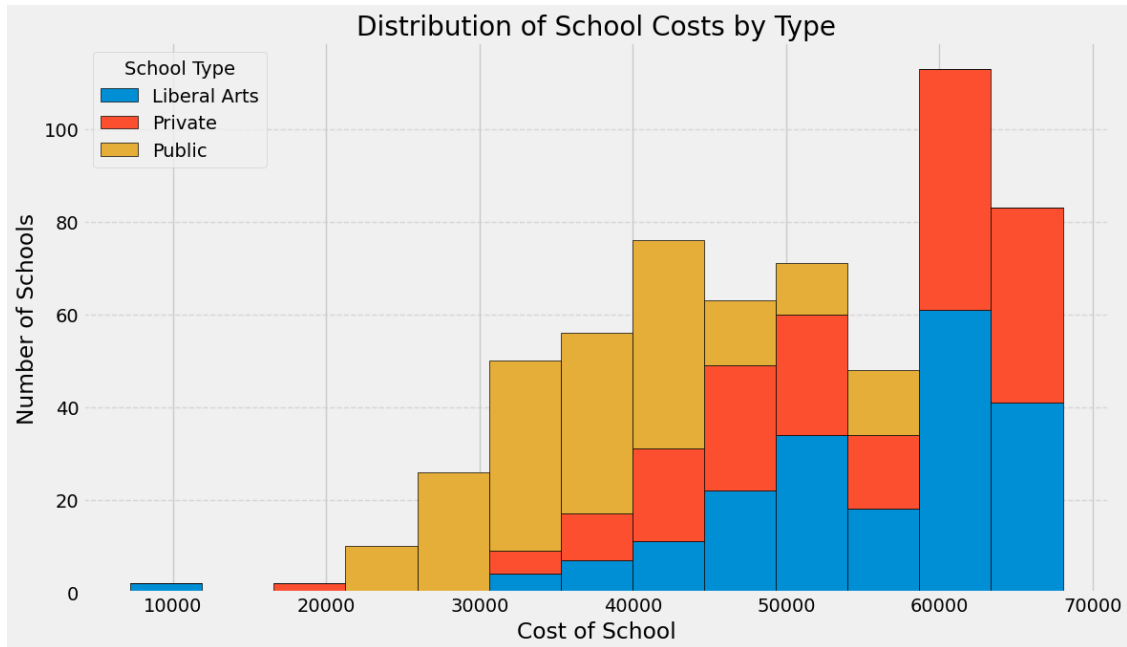
plt.hist(costs_by_type, bins='auto', stacked=True, label=school_types,
↳edgecolor='black')

# Add titles and labels for clarity
plt.title('Distribution of School Costs by Type')
plt.xlabel('Cost of School')
plt.ylabel('Number of Schools')

# Add a legend to explain what each color/stack represents
plt.legend(title='School Type')

# Add a subtle grid for better readability
plt.grid(axis='y', linestyle='--', alpha=0.7)

# Make sure everything fits nicely in the plot area
plt.tight_layout()
```



1.5 Problem 5

In the cell below make boxplots of the height variable in the Height_Weight_Age_Sex dataset, separated by sex.

```
[75]: # imports
import matplotlib.pyplot as plt
import pandas as pd

# Since we do not have access to Height_Weight_Age_Sex to call, we'll create
↳ our own mini database
data = [
    {"Height (cm)": 162, "Weight (kg)": 55, "Age (years)": 25, "Sex": "Female"},
    {"Height (cm)": 168, "Weight (kg)": 60, "Age (years)": 30, "Sex": "Female"},
    {"Height (cm)": 155, "Weight (kg)": 50, "Age (years)": 20, "Sex": "Female"},
    {"Height (cm)": 170, "Weight (kg)": 68, "Age (years)": 35, "Sex": "Female"},
    {"Height (cm)": 160, "Weight (kg)": 54, "Age (years)": 28, "Sex": "Female"},
    {"Height (cm)": 165, "Weight (kg)": 57, "Age (years)": 23, "Sex": "Female"},
    {"Height (cm)": 158, "Weight (kg)": 52, "Age (years)": 27, "Sex": "Female"},
    {"Height (cm)": 175, "Weight (kg)": 75, "Age (years)": 28, "Sex": "Male"},
    {"Height (cm)": 180, "Weight (kg)": 80, "Age (years)": 35, "Sex": "Male"},
    {"Height (cm)": 170, "Weight (kg)": 72, "Age (years)": 25, "Sex": "Male"},
    {"Height (cm)": 185, "Weight (kg)": 85, "Age (years)": 30, "Sex": "Male"},
    {"Height (cm)": 178, "Weight (kg)": 78, "Age (years)": 32, "Sex": "Male"},
    {"Height (cm)": 172, "Weight (kg)": 70, "Age (years)": 29, "Sex": "Male"},
]
```

```

    {"Height (cm)": 182, "Weight (kg)": 82, "Age (years)": 27, "Sex": "Male"}
]

# Convert data to a Pandas DataFrame
df = pd.DataFrame(data)

# Create separate data for each sex
female_heights = df[df["Sex"] == "Female"]["Height (cm)"]
male_heights = df[df["Sex"] == "Male"]["Height (cm)"]

# Create boxplots
plt.figure(figsize=(8, 6))
plt.boxplot([female_heights, male_heights], labels=["Female", "Male"],
            patch_artist=True,
            boxprops=dict(facecolor="lightblue", color="blue"),
            medianprops=dict(color="red"))
plt.title("Boxplot of Heights by Sex")
plt.ylabel("Height (cm)")
plt.xlabel("Sex")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.show()

```

/var/folders/wb/pw5392695b13z5sd18v31vgh0000gn/T/ipykernel_31216/3171041264.py:3
 3: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been
 renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be
 dropped in 3.11.

```

plt.boxplot([female_heights, male_heights], labels=["Female", "Male"],
            patch_artist=True,

```

